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TASK 3 – Design the Control Logic

Marks: 4

This task covers two important operations in the Smart-Parking System:

1. Allocating an available parking slot.
2. Calculating the parking fee.

The assignment also requires pseudocode and a completed Java program. CO1 JAVA

1. Slot Allocation Logic

Slot allocation means finding an empty parking slot and assigning a vehicle to it.

Working:

1. Start searching from the first floor.
2. Check each parking slot on that floor.
3. If a slot is empty, assign the vehicle to it.
4. Return the floor number and slot number.
5. If all slots are occupied, return null.

Pseudocode:

```
ALGORITHM AllocateSlot(vehicleNo)


BEGIN
  FOR each floor f
    FOR each slot s in floor f


      IF slot is empty THEN
        Assign vehicleNo to slot[f][s]
        RETURN {f, s}
      END IF


    END FOR
```

```
    END FOR

    RETURN NULL

END
```

2. Parking Fee Calculation Logic

The parking fee depends on the vehicle's parking duration.

The system can use the following fee rules:

- First hour: First-hour charge.
- Each additional started hour: Additional-hour charge.
- Daily maximum: The fee cannot exceed the daily maximum for a 24-hour period.

Example: Assume the first-hour charge is Rs. 30, each additional hour costs Rs. 20, and the daily maximum is Rs. 200.

Parking duration	**Billable hours**	**Fee**
30 minutes	1	Rs. 30
60 minutes	1	Rs. 30
90 minutes	2	Rs. 50
120 minutes	2	Rs. 50
180 minutes	3	Rs. 70

These are illustrative rates because the assignment does not specify actual parking charges.

Pseudocode:

```
ALGORITHM CalculateFee(minutes)

BEGIN
    IF minutes <= 0 THEN
        RETURN 0
    END IF
```

```
    fullDays = minutes / 1440
    remainingMinutes = minutes MOD 1440

    fee = fullDays * dailyMaximum

    IF remainingMinutes > 0 THEN

        hours = (remainingMinutes + 59) / 60

        IF hours == 1 THEN
            partialFee = firstHourRate
        ELSE
            partialFee = firstHourRate
                    + (hours - 1) * additionalHourRate
        END IF

        IF partialFee > dailyMaximum THEN
            partialFee = dailyMaximum
        END IF

        fee = fee + partialFee

    END IF

    RETURN fee

END
```

Here, 1440 represents the number of minutes in 24 hours.

The calculation uses integer arithmetic to charge every started hour fully.

3. Use of break, continue and return

These statements control the execution of loops and methods.

break: Stops the current loop immediately. It can be used when the required parking slot is found.

continue: Skips the current iteration and moves to the next iteration. For example, when a slot is already occupied, the program checks the next slot.

return: Immediately exits a method and sends a value back. In slot allocation, it returns the allocated floor and slot numbers. If no slot is available, it returns null.

When no suitable parking slot is available, the system displays that parking is full